

Abhishek Dhanani

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EDUCATION

Sacred Heart University

Master's in computer science – data science track **GPA: 3.9**

- Coursework: Machine learning, Deep Learning, Big Data Analysis

Fairfield, CT

Aug. 2024 – Dec. 2025

Government Engineering College

Bachelor's in computer science

- Coursework: Algorithms, Database Management, Advanced Mathematics, Software Engineering

Gandhinagar, Gujarat, India

Aug. 2020 – May 2023

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java

Data Visualization: Tableau, Matplotlib, Seaborn

Developer Tools: Git, Visual Studio, PyCharm, Jupyter notebook

Machine Learning Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers

EXPERIENCE

Data analysis intern

Infolabz Pvt. Ltd.

Jan 2023 – July 2023

Ahmedabad, INDIA

- Cleaned and analyzed data using Pandas and NumPy to identify trends, patterns and actionable insights using tools like excel and Python.
- Developed visualizations and reports with Tableau and Matplotlib to present findings to teams.
- Contributed to the development of regression and classification models with Scikit-learn, to gain meaningful insights about data and analyze the market trends.
- Worked closely with marketing team to gather data requirements and deliver insights that supported strategic decision-making resulting in sales improvement.

PROJECTS

[NLP Exploration of United Nations Discourse](#) | NLP, NLTK, spaCy, Python, LDA, NFA

Dec 2025

- Analyzed over 4,700 UN General Debate speeches (2000-2024) using **NLP techniques** for tokenization, word frequency analysis, sentiment analysis, and **topic modeling (LDA/NMF)** to uncover trends in global discourse on climate, security, and development.
- Generated visualizations including heatmaps, word clouds, and trend charts to highlight shifts (e.g., increased focus on sustainability post-2015 and regional variations across G7, G20, EU, and Africa).
- Paper accepted** to the 2026 International Conference on Breakthroughs in Artificial Intelligence (**BAI'26**); to be published in Springer Lecture Notes in Networks and Systems (LNNS).
- Demonstrated expertise in data preprocessing, machine learning frameworks, vector databases, and interactive tools (pyLDAvis).

[MailCast: Email to Audio Generation](#) | NLP, Python, PyDub, OpenAI API, Gemini API

Mar 2025

- Developed a **Python application** to convert emails into audio podcasts by integrating **Gmail API** for email retrieval and BeautifulSoup for **text preprocessing**.
- Utilized a large language model (**LLM**) to generate podcast scripts from email content and **OpenAI's Text-to-Speech (TTS) API** with **PyDub** for audio output.
- Implemented **OAuth2 authentication** and chunked **text processing** to handle API token limits, ensuring scalability for large email datasets.
- Demonstrated skills in API integration, text processing, natural language generation, and audio generation.

[RAG System for Document Summarization](#) | Python, LangChain, ChromaDB, OpenAI Embedding, GPT models

Mar 2025

- Designed a Retrieval-Augmented Generation (**RAG**) system in Python to summarize lengthy legal documents, preprocessing text and storing summaries in a **Chroma vector database**.
- Integrated **OpenAI embeddings** and GPT models with **LangChain** to retrieve and generate concise, accurate summaries from document sections.
- Built a **multi-vector retriever pipeline** to combine original text and summaries, enhancing retrieval efficiency and summary coherence.
- Showcased expertise in NLP, vector databases, retrieval techniques, and large language models.

- Developed an **NLP model** to classify news articles as real or fake by **fine-tuning DistilBERT** using **Hugging Face's Transformers** library and **AutoTokenizer** for efficient text preprocessing.
- Processed a dataset of over **44,000 articles** (Fake.csv and True.csv), performing data cleaning, tokenization, and splitting (70% train, 15% validation, 15% test).
- Fine-tuned DistilBERT on **PyTorch** with GPU, leveraging mixed precision training (FP16) for faster computation and reduced memory usage. Achieving an accuracy of 99.98% on the test set, with a confusion matrix and classification report for evaluation.
- Demonstrated skills in NLP, deep learning, model optimization, and Hugging Face ecosystem integration.